NAME

mbusbl2fnv − converts a USBL (ultra-short baseline) ROV tracking CSV file to an MB-System fast navigation (fnv) text file.

VERSION

Version 5.0

SYNOPSIS

mbusbl2fnv [**−I***usbl.csv*] [**−O***output.fnv*] [**−V**] [**−H**]
mbusbl2fnv [**−−input**=*usbl.csv*] [**−−output**=*output.fnv*] [**−−verbose**] [**−−help**]

DESCRIPTION

mbusbl2fnv converts a USBL (ultra-short baseline) ROV tracking CSV file to an MB-System fast navigation (fnv) ASCII text file.

The input CSV file is expected to have one header line followed by data lines, each containing four comma-separated fields:

field 0: epoch time (Unix epoch seconds)
field 1: latitude (decimal degrees, positive north)
field 2: longitude (decimal degrees, positive east)
field 3: ROV depth (meters, positive down) − used as draft

Heading, speed, roll, pitch, and heave are not present in the USBL tracking data and are output as 0.0 in the fnv file.

Lines are treated as data only if their first non-whitespace character could begin a number (a digit, **+**, **−**, or **.**); this allows the header line and any comment lines beginning with **#** to be skipped automatically. Lines that do not parse into exactly four numeric fields are skipped with a warning in verbose mode. Records with retrograde (non-increasing) timestamps are also skipped.

The fnv format is a tab-separated ASCII text file. The first line is a comment beginning with **##** that names the columns. Each subsequent data line contains the following tab-separated fields:

yyyy mm dd hh mm ss.ssssss
epoch_seconds
longitude (deg)
latitude (deg)
heading (deg)
speed (km/hr)
draft (m)
roll (deg)
pitch (deg)
heave (m)

The date and time in the first field are space-delimited within their tab field.

If no input file is specified, **mbusbl2fnv** reads from standard input. If no output file is specified, it writes to standard output.

MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute

OPTIONS

−H

−−help

Prints the program name, usage, and a description of the expected USBL CSV field layout, then exits.

−V

−−verbose

Increases the verbosity of **mbusbl2fnv**. With this option the program prints the input and output file names to standard error when they are opened, reports any malformed or retrograde-timestamp lines that are skipped, and prints a final summary of the number of records read, written, and skipped.

−I *usbl.csv*

−−input=*usbl.csv*

Sets the path of the input USBL tracking CSV file. If this option is not given, the program reads from standard input.

−O *output.fnv*

−−output=*output.fnv*

Sets the path of the output fnv file. If this option is not given, the program writes to standard output.

EXAMPLES

Convert a USBL tracking CSV file to an fnv file:

```
mbusbl2fnv −I usbl_track.csv −O rov.fnv
```

Use standard input and output (e.g. within a pipeline):

```
cat usbl_track.csv | mbusbl2fnv > rov.fnv
```

Convert with verbose progress reporting:

```
mbusbl2fnv −−verbose −−input=usbl_track.csv −−output=rov.fnv
```

Use the fnv output to supply ROV navigation to **mbpreprocess**:

```
mbusbl2fnv −I usbl_track.csv −O rov.fnv
mbpreprocess −−format=<format> −−input=datalist.mb−1 \
   −−nav−file=rov.fnv −−nav−file−format=9 \
   −−sensordepth−file=rov.fnv −−sensordepth−file−format=9
```

SEE ALSO

mbsystem(1), **mbnavlab2fnv**(1), **mbfnv2navlab**(1), **mbpreprocess**(1), **mbprocess**(1)

BUGS

Heading, speed, roll, pitch, and heave are not available in USBL tracking data and are always written as 0.0 in the fnv output. Applications that require these values should obtain them from another navigation or attitude source.

The optional port and starboard swath edge columns present in some fnv files are not written by **mbusbl2fnv**, as no swath geometry is available from USBL tracking data.